

1 Griffith's Problem 2.2

Lemma 1.1 (Convexity). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a smooth¹ function such that f'' , the second derivative of f , is non-negative. Then, for any real numbers $a < x < b$,*

$$f(x)(b-a) \leq (b-x)f(a) + (x-a)f(b). \quad (1)$$

*This states that f is a **convex** function.*

Proof. Let f be a smooth function possessing a non-negative second derivative, and let $a < b$ be real numbers. Denote $g_{ab}(x) = (b-x)f(a) + (x-a)f(b)$, proportional to the secant line between $f(a)$ and $f(b)$. Consider the difference² between f and its secant line:

$$\begin{aligned} g_{ab}(x) - f(x)(b-a) &= g_{ab}(x) - (x-a)f(x) - (b-x)f(x) \\ &= (x-a)\{f(b) - f(x)\} - (b-x)\{f(x) - f(a)\} \\ &= (x-a)(b-x) \frac{f(b) - f(x)}{b-x} - (b-x)(x-a) \frac{f(x) - f(a)}{x-a} \\ &= (x-a)(b-x) \left(\frac{f(b) - f(x)}{b-x} - \frac{f(x) - f(a)}{x-a} \right) \end{aligned}$$

Apply the **Mean Value Theorem** (MVT) to f , feasible by virtue of its differentiability. As such, we are guaranteed the existence of $c_a \in [a, x]$ and $c_b \in [x, b]$, such that

$$f'(c_a) = \frac{f(x) - f(a)}{x-a}, \quad f'(c_b) = \frac{f(b) - f(x)}{b-x} \quad (2)$$

We may apply the MVT once more within the interval $[c_a, c_b]$, this time on f' , since f is (at least) twice differentiable. The theorem guarantees a $d \in [c_a, c_b]$ such that

$$f'(c_b) - f'(c_a) = (c_b - c_a)f''(d) \quad (3)$$

Applying equations (2) and (3), one arrives at the condition

$$g_{ab} - f(x)(b-a) = (x-a)(b-x)(c_b - c_a)f''(d) \geq 0, \quad (4)$$

where the last inequality arises from the assumption of a non-negative second derivative, along with the fact that $a < c_a < x < c_b < b$. The proof may be trivially concluded here. $\square$

Lemma 1.2. *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. If g is bounded from above³, it must be constant.*

Proof. For a contradiction, assume that g is convex, bounded from above, and non-constant. Thus, there must be some $c, d \in \mathbb{R}$ such that $g(c) < g(d)$. Assume,

¹ $f \in C^2(\mathbb{R})$ suffices.

² To be more specific, consider the difference scaled by $(b-a)$.

³ There is some $M \in \mathbb{R}$ such that $g(x) < M$ for any $x \in \mathbb{R}$.

without loss of generality that $c < d$ (one could always play similar tricks with $g(-x)$). Employing the statement of convexity (1),

$$g(x) \geq g(c) + (x - c) \frac{g(d) - g(c)}{d - c}, \quad x > d \quad (5)$$

Clearly, since x can be arbitrarily large, we have arrived at a contradiction to the boundedness of g . Thus, if g is bounded above and convex, it must be constant. $\square$

Theorem 1.3. *Consider a normalisable real solution to the time-independent Schrodinger equation provided by some potential⁴ $V(x)$. The energy of such a solution is bounded below by the minimum of the potential.*

Proof. Assume that ψ is a real solution to the time-independent Schrodinger equation provided by some potential $V(x)$, with energy E . Further, assume that $V(x) > E$ for any value of x , and that ψ is normalisable. A rearrangement of the Schrodinger equation provides

$$\psi''(x) = \frac{2m}{\hbar^2} (V(x) - E) \psi(x).$$

The assumption of $V(x) > E$ is then identical to the statement that $\psi''(x)$ and $\psi(x)$ share the same sign. Consider $|\psi|^2 = \psi^2$. Our assumption renders its second derivative non-negative. This is rendered obvious by a short calculation:

$$\frac{d^2}{dx^2} \psi^2(x) = 2(\psi'(x)^2 + \psi(x)\psi''(x)) \geq 0,$$

where the last inequality arises from the facts that ψ and ψ'' share the same sign, making their product non-negative, while the *square* of a function $\psi'(x)^2$ trivially shares this property. By Lemma 1.1, ψ^2 is convex.

Either ψ^2 is bounded from above, or it isn't.

Consider the case of an unbounded ψ^2 . The proof of Lemma 1.2 provides the lower bound (5), implying that if $x > x_0$, $\psi^2(x) > M_0$ for some $x_0, M_0 \in \mathbb{R}$. Since ψ^2 is always non-negative, and is non-zero at some point (otherwise, it would be bounded), $M_0 > 0$ may be chosen by employing an appropriate choice of x_0 . The non-negativity of ψ^2 further implies that for any real numbers $a < b$,

$$\int_a^b \psi^2(x) dx \geq 0.$$

⁴Technically, we assume that the potential is bounded below, but a failure to assume this would render the theory unstable against decay.

Then,

$$\begin{aligned}\int_{-\infty}^{\infty} \psi^2(x) dx &= \int_{-\infty}^{x_0} \psi^2(x) dx + \int_{x_0}^{\infty} \psi^2(x) dx \\ &\geq \int_{x_0}^{\infty} \psi^2(x) dx \\ &\geq M_0 \int_{x_0}^{\infty} dx.\end{aligned}$$

Thus, the norm of ψ^2 diverges if it is unbounded, contradicting its normalisability.

We are only left with a single option: ψ^2 is bounded from above. Lemma 1.2 immediately asserts that ψ^2 must be constant. It is trivially observable that the only constant function that may be integrated over the entire set of real numbers⁵ to produce a finite result is the function $\psi^2(x) = 0$. However, this implies that ψ has a vanishing norm, and is therefore not normalisable.

This concludes the proof of the contra-positive: if $V(x) > E$ for all positions $x \in \mathbb{R}$, ψ is not normalisable.

□

⁵That is, $\int_{-\infty}^{\infty}$.